UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

General Certificate of Education O Level

MARK SCHEME for the November 2004 question paper

5090 BIOLOGY

5090/02 Paper 2 (Theory), maximum mark 80

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UNIVERSITY of CAMBRIDGE
International Examinations

NOVEMBER 2004

GCE O Level

MARK SCHEME

MAXIMUM MARK: 80

SYLLABUS/COMPONENT: 5090/02

BIOLOGY
Paper 2 (Theory)

UNIVERSITY of CAMBRIDGE
International Examinations

Section A

1 (a) **A** - guard cell ;

B - epidermis/al cell (**R** lower epidermis) ;

C - phloem/sieve tube (**A** companion) ; **3**

(b) (i) allows leaf to float AW/(maximum) exposure to light* ;

(**R** support unqualified)

(ii) diffusion/movement/collection/source/provides/gives AW + CO_2

OR (maximum) exposure to light* (*once only)

(Ignore references to oxygen, but **R** O_2 references if they refer to respiration) ; **2**

(**R** absorbs/takes in/references gas exchange)

(c) (Ignore references to leaf stalks and to spaces not interconnected)

stomata/guard cells (mainly) on upper surface AW ;

(or v.v.)

air spaces/chambers + palisade cells (or pos^{n} described) ;

chloroplasts/chlorophyll in epidermis (**R** upper epidermis) ;

reference cells in clumps v. cells loosely packed AW/
air chambers v. intercellular spaces/
large spaces v. small spaces (**R** more/fewer spaces) ;

no cuticle on lower surface ;

reference quantity of chloroplasts/chlorophyll in spongy cells ;

max. 3

(d) less/no + thickening/lignin/xylem/woody (or v.v.) ;

(R unqualified references to hard/rigid)

no need for support/support from water (or v.v.) ; **2**

(**A** floats on)

Total 10

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2 (a) large(r) diameter at low light intensity/or v.v. ;

(**A** bigger/inversely proportional or description) (**R** proportional unqualified)

fastest rate of change around 2 - 4 a.u./
slowest rate of change/levels off at 7 - 10 a.u. ; **2**

(b) reflex/autonomic/automatic/involuntary ; **1**

(**R** spinal/conditioned)

(c) light sensitive/receptor (cells) or named/retina ;

neurones/nerve cells or fibres (**A** optic nerve) ;

impulses ;

contraction + circular muscles (**R** if reference ciliary) ;

relaxation + radial muscles (**R** if reference ciliary) ;

correct reference iris ;
max. 5

(d) no colour/pigment in iris/choroid (**R** eye) ;

permits internal reflection AW of light/too much light enters
eye/received by retina (**A** no shading/shielding/protection for retina) ;

damage to retina/receptors/light-sensitive + cells/visual impairment AW
(**R** damage to eyes) ; **3**

Total 11

3 (a) one chromosome shown - in a string (mark the first) ;

genes matching in shape and sequence (**A** reversed) ;
(the appropriate 4 may be selected from a string of more than 4)

gene 3 not shaded (all others must be uniform black or white) ; **3**
(gene 2 if the chromosome has been reversed)

(b) (i) mutation (ignore reference chromosome) ; **1**

(ii) mutagen (or named)/reference change in DNA structure ; **1**

(**A** any plausible e.g. - radiation or named (α-/γ-/X-rays)/chemicals
/u.v./sunlight/carcinogens/smoking/viruses) ;

(**R** heat/infra-red/disease)

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(c) (i) $\underline{I^A}$;

$\underline{I^o}$ (allow in either order) ; **2**

(ii) O/I° from partner/offspring must be I° I° or OO ;

$\underline{A/I^A \text{ or } B/I^B}$ from the person/person cannot supply I°/O ;
(must have reference to both alleles)

I^A and I^B are dominant* (to I°) / I° recessive* (to both) ; **3**
(*AW) (**A** references to A/B/O without I)

Total 10

4 (a) ecosystem ; **1**

(**A** light/sun)

(b) energy entering producer/plant/tree/leaf (**A** no arrow head) ;

(**R** unlabelled arrow) (**A** unlabelled drawings)

plant/tree/leaf ➔caterpillar ➔ bird (arrows must be present) ; **2**
(and in correct direction)

(**R** tree ➔ leaf)

(c) (i) correct pyramidal shape (**A** inverted pyramid) ;

all levels correctly identified with labels (**A** tree + leaf here) ; **2**

(tree will be on top if inverted but **R** producers/consumers as labels)

(ii) bottom or top block smallest and labelled tree AW ;
or largest and labelled leaf

working away from the tree/leaf -
other two blocks large then small + correctly labelled ; **2**

(d) block of fleas/parasites larger than and next to birds ;

rest of pyramid a reasonable copy of that in **(c) (ii)** (**A** e.c.f.) ; **2**

(unless **(c) (ii)** is wrong and **(d)** is correct)

Total 9

5 (a) G oesophagus/gullet ;

H stomach ;

I colon/large intestine/large bowel ; **3**

(b) $\underline{\text{E/ileum}}$ (**R** small intestine) ; **1**

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(c) (i) 2 h(ours)/120 minutes (units required) ; **1**

(ii) stomach/**H** ; **1**

(d) acid resistant coat (**R** in BI context) ;

not affected by HCl/acid in <u>stomach</u> ;

drug not released until duodenum/small intestine AW/leaves stomach/meets alkaline environment (**A** letters) ;

takes longer for water to enter/drug to dissolve ;

membrane slows down speed of drug release ;

max. 3

(e) reference sticks to mucus + in intestine AW (**R** oesophagus/stomach) ; **1**

Total 10

Total mark for Section A = 50

Section B

6 (a) correct reference atria(um)/auricle(s) ;

correct reference ventricle(s) ;

<u>muscles/muscular</u> + <u>contract(ion)</u> (**R** pushing/forcing pumping - in Q.) ;

reference thickness of ventricular compared with atrial walls ;

atrio-ventricular/identified valve(s) (open) + blood passes ;

close + to prevent return of blood ;

tendons/cords/(**R** heartstrings) + action/function of ;

reference aortic valves + their action (**A** close prevent backtflow) ;

cycle repeated/idea of co-ordinated action; ;

max. 7

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(b) right (ventricle) wall thinner/left (ventricle) wall thicker OR reference less/ more muscle OR weaker/stronger contractions ;

(**A** smaller—Larger)

(pulmonary) shorter distance to travel (**A** only to the lungs) (or v.v.) ;

little work to do against gravity (the idea of) (or v.v.) ;

avoidance of damage to lung <u>capillaries</u>/low pressure required in lungs ;

(body) high pressure for kidney filtration ;

oxygen/glucose to brain ;

max. 3

Total 10

7 **(a)** anywhere – one correct reference stomatal movement + effect ;

– (ignore references to water vapour)

(i) dark/no light + no photosynthesis ;

(**R** night)

respiration occurring ;

*CO_2 out/released/produced + O_2 in/absorbed/used ;

(ii) light/day + photosynthesis ;

faster than respiration AW ;

*O_2 out/released/produced + CO_2 in/absorbed/used ;

max. 5

(* accept on **<u>annotated</u>** equation)

(b) **(i)** reference concentration gradients of CO_2/O_2 ;

CO_2 is a limiting factor/the more CO_2 the faster the P/S ;

more or faster CO_2 in + more or faster O_2 out ;

(ii) wilting/cells flaccid AW (R plasmolysis) ;

stomata close ;

slower exchange of gases (**R** no exchange) ;

slower rate of P/S (**R** no P/S) ;

max. 5

Total 10

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8 Either (a) (i) sperms + ova/eggs [anywhere in **(a)**] ;

smaller/larger/correct size reference of either

(ova – 120 to 150μm, sperm 60μm with head diameter 2.5μm x 3μm) ;

many can be released/sperm is only nucleus + tail

OR ovum carries some nutrition/cytoplasm/yolk (or v.v.) ;

sperm small enough to enter egg ;

(ii) ratio – large numbers : one/few (A lifetime numbers) ;

(A 1 000 minimum)

greater wastage/chance of fertilisation/sperms

(**A** more die) reaching ovum ;

limited space for embryo/fetus/baby/room only for a few embryos/ fetuses/babies ;

fixed number of eggs (ova)/ova present from birth/sperms produced continuously ;

(iii) sperms have tail/flagellum/swim/motile (**R** move) ;

to reach egg/ovum/reference fertilisation + in oviduct ;

(**A** Fallopian tube)

ova experience only passive movement (or described) ;

max. 8

(b) (i) copulation AW + when no ovum in system/at infertile time/stated time in cycle (**A** any time outside 5 days before ovulation to 7 days after)/#withdrawal method explained/*abstinence1 ; **1**

(**R** rhythm method unqualified)

(ii) (linked to **(i)** above, but can score if **(i)** is left blank)

cycle variable or irregular/description of irregularity/miscalculation/ misinterpretation of raised temperature/

#some sperms released before ejaculation/

*lack of control – (BUT **A** this IS the safest method) ; **1**

(if they say it)

Total 10

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8 OR (a) (i) (**female**) one per ovule ;

comparatively few ovules/gametes (per plant or flower) ;

parent must supply space/food for developing seed ;

(**male**) millions/lots of male gametes/pollen (grains) ;

(**A** 1 000 minimum)

great wastage/many may die/pollination is very chancy ;

(ii) female gamete does not move/is attached to ovule/ovary ;

already positioned where it will develop AW ;

male gamete/pollen is moved by named agent ;

gamete is inside pollen grain ;

described adaptation of pollen grain for dispersal ;

to carpel/stigma ;

then moves within/by growth of the pollen tube ;

max. 7

(b) same (properties) as parent/genetically identical AW ;

only one parent needed/no need for gametes/no agents needed/ faster ;

less wastage/more certain ;

offspring bound to be in suitable environment AW ;

well-developed before separation from parent/allows (rapid) colonisation ;

max. 3

Total 10

Total mark for Section B = 30

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